ClipForge — AI Stream Clipping Platform Blueprint

Oct 17, 2026

1. Executive summary and the decision

Build a rights-first clipping tool for **Twitch streamers and the editors they authorize**, launched as a modular monolith (Next.js + FastAPI + Postgres + Redis queue + FFmpeg workers), with uploads as the universal ingestion path and Twitch's official clip APIs as the first integration. "ClipForge" is a working name.

The market is crowded (OpusClip, Vizard, Klap, StreamLadder, Eklipse, Powder, Submagic) and general "long video → shorts" is a commodity priced at roughly $15–$29/month. The defensible wedge is narrower: **stream-native highlight detection (chat + audio + transcript), a creator-authorized editor workflow, and a per-creator feedback loop.** Generic podcast clipping is not where we win.

Three platform facts shape everything (verified 17 Oct 2026, details in section 2):

- **No major source platform lets a third party download full VODs through an official API.** Twitch offers clip creation from VODs and clip downloads for broadcasters/editors; Kick's official API has no VOD endpoints; YouTube's Data API never returns media files. Uploads by the rights-holder are the reliable path.
- **Every publishing API gates public posting behind an audit.** Unaudited TikTok and YouTube apps can only post privately. Audits must start in week 1, not after launch.
- **X is now pay-per-use** and Instagram/Facebook publishing requires professional accounts. Publishing is a feature with per-platform caveats, not a uniform button.

Strategic decision (section 17 answered up front)

#	Question	Answer
1	What v1 contains	Upload + Twitch-authorized import; transcription; multi-signal highlight ranking; candidate review; trim editor; vertical reframe (face-aware + split-screen); animated burned-in captions; render + download; YouTube Shorts and TikTok publishing (once audited); usage-metered billing.
2	Deliberately excluded	Live in-stream clipping, Kick automated import, Instagram/Facebook/X publishing, agency white-label, template marketplace, analytics feedback loop, natural-language search (phase 5), mobile editor (view/approve only).
3	First integration	**Twitch.** It is the only source with official, sanctioned clip creation from VODs and clip download for editors, plus chat data (via EventSub/IRC while live) that competitors serving podcasts lack.
4	AI worth paying for	Hosted batch speech-to-text (≈$0.15–$0.26/hour of audio) and a small/medium LLM for clip titling and context-completeness scoring. Everything else starts open-source.
5	Open-source	FFmpeg, PySceneDetect, MediaPipe/face detection, audio event models (PANNs/YAMNet), pgvector, Whisper (as fallback/self-host once volume justifies a GPU).
6	Minimum launch budget	**≈ $6k–$15k cash** excluding founder salaries (legal review $2–5k, infra and AI during build and beta ≈ $1.5–3k, design/brand $1–3k, buffer). With paid contractors add $40k–$90k. Estimate, not quote.
7	Monthly operating cost at launch	**≈ $400–$1,200/month** for ⩽ 300 active users (section 22).
8	Primary revenue tier	**Pro ($29/mo)** — streamers posting daily who need publishing and higher limits. Agency is the margin expansion, not the launch driver.

#	Question	Answer
9	Differentiation	Chat-aware stream highlights, editor authorization built in (a streamer grants an editor rights in one click), and per-creator learning from actual post performance.
10	Biggest risks	Commercial: commodity pricing and incumbents copying stream features. Technical: highlight quality on gameplay with little speech. Platform: audit denials, API policy changes, Twitch Clip-from-VOD being beta.
11	Validate first	20 streamer/editor interviews; a concierge test (manually clip 5 streamers' VODs with scripts for 2 weeks); price test at $19/$29; whether streamers will grant editor access.
12	First 30 days	Week 1: audits submitted, upload → transcript → candidates script. Week 2: scoring spike measured against hand-labeled highlights. Week 3: editor + render. Week 4: billing, Twitch OAuth, closed beta with 10 creators. Detail in section 24.

Biggest honest caveat: highlight scoring predicts *clip-worthiness*, never views. The product must say so in its UI.

2. Market, competitors and platform reality

Competitors converge on per-minute-of-source pricing; the opening is stream-specific intelligence and editor workflows, not cheaper clipping.

2.1 Competitor landscape

Product	Focus	Pricing signal	Gap we exploit
OpusClip	Podcast/talking-head to shorts	Free; Starter $15/mo; Pro $29/mo ($174/yr); Business custom. 1 credit = 1 source minute	Weak on gameplay with little speech; no chat signal

Product	Focus	Pricing signal	Gap we exploit
Vizard, Klap, Submagic	Generic repurposing, captions	Similar $15–$40 band (verify before pricing)	Same commodity feature set
StreamLadder, Eklipse, Powder	Gaming/stream clips	Freemium + subscription (verify)	Shallow editor-team and rights workflows
Manual editors (Premiere, CapCut)	Full control	Labor cost dominates	Hours of VOD review

Common complaints to validate in interviews (assumptions, not verified data): clips cut mid-sentence, poor gameplay detection, credit burn on long VODs, unreliable auto-publishing, no way to manage many creators.

2.2 Source platforms (what the official APIs actually allow)

Platform	Official capability	Media file?	Live?	Owner auth needed	Our workflow
Twitch	Helix API: Get Videos (metadata), Create Clip, Get Clips, Get Clips Download (broadcaster/editor), Create Clip From VOD (open beta)	Full VOD: **no**. Clips: **yes** for broadcaster/editor	Clip from earlier in current stream if VODs enabled	Yes (broadcaster or editor)	Detect moments from chat/metadata → create Twitch clip → download → edit. Full VOD by owner upload
Kick	Official public API (api.kick.com/public/v1): channels, chat, livestreams, events, OAuth 2.1 + PKCE	**No VOD endpoints.** VOD data only via undocumented website endpoints that may break	Metadata/chat only	Yes	Upload only. Do not build on undocumented endpoints
YouTube	Data API v3: video metadata, captions (owner), uploads	**No** media download	No ingest	Yes	Owner uploads file (or Google Drive import). Read metadata via API

Twitch staff state that VOD downloaders violate the Developer Agreement (forum). The design never scrapes, never strips DRM, never uses yt-dlp against third-party content.

2.3 Publishing platforms

Destination	Direct publish	Account type	Gate	Limits (verified)
TikTok Content Posting API	Direct Post or Upload-to-inbox (draft)	Any, via Login Kit	Unaudited: posts forced private, ⩽ 5 users/24 h, accounts must be private. Audit required	~15 posts/creator/day shared across all apps; must query creator_info before each post; commercial content disclosure UI required
YouTube Shorts (videos.insert)	Yes; no Shorts flag (⩽ 3 min vertical = Short)	Channel owner	Unaudited projects created after 28 Jul 2020: uploads locked private. Compliance audit	10,000 units/day default; upload quota reported moved to its own bucket (~100 calls/day) in 2026 — **verify in Cloud Console**
Instagram Reels Content Publishing	Yes, two-step container flow	Professional (Business/Creator) only	Meta App Review; Page Publishing Authorization where required	Docs state both 50 and 100 posts/24 h — enforce 50 and query content_publishing_limit; containers expire in 24 h
Facebook Reels	Yes (Pages)	Page	App Review	Verify per endpoint
X	Yes	Any	Pay-per-use credits since 6 Feb 2026	≈$0.015/post, ≈$0.20 with a link; 100 posts/15 min/user (third-party reports — verify docs.x.com)

Fallback everywhere: download the file, copy generated caption/hashtags, deep-link to the app.

3. Product requirements and workflows

The core loop is import → candidates in under 15 minutes for a 2-hour VOD → edit → render → download or publish.

3.1 Personas and jobs

Persona	Primary job	Launch priority
Streamer	Turn own VOD into daily shorts with minimal effort	P0
Clip editor	Produce clips for 1–10 streamers who authorized them	P0
Creator manager	Approve and schedule across creators	P2
Clipping agency	Volume, client workspaces, billing	P2
Fan / community	Clip with permission only (Twitch's native Clip rights; creator-run clip programs)	P3
Content business	Repurpose owned webinars/podcasts	P3 (opportunistic)

3.2 Authentication and onboarding

1. Sign up (email+password with verification, or Google/Twitch OAuth). Argon2id hashing; optional TOTP 2FA; passkeys later.
2. Role question: "I clip my own streams" / "I clip for others" / "Team".
3. Defaults: clip length (15–60 s), aspect (9:16), caption style, language.
4. Connect Twitch (optional, recommended). Scopes: `clips:edit` (create clips), `editor:manage:clips` / `channel:manage:clips` for downloads (confirm exact scope names in the Helix reference before build), `user:read:email`.
5. Rights attestation: "I own this content or have the creator's written permission." Stored with timestamp, IP, version of terms.
6. First import with a sample video if they skip.

Declining Twitch connection: upload still works fully. Declining social connections: download + copy-caption flow.

3.3 Ingestion workflows

Path	How	Rights check	Notes
Local upload	Resumable multipart to S3-compatible storage (tus or S3 multipart, 64 MB parts)	Attestation per asset	Primary path; up to 12 h / 20 GB on paid tiers
Cloud import	Google Drive / Dropbox picker (user OAuth)	Attestation	Server pulls the file with user token
Twitch authorized	User is broadcaster or editor → list VODs (metadata) → our detector proposes moments → Create Clip From VOD → Get Clips Download	Twitch role verified via API	Clip-level media only; beta endpoint, feature-flagged
Twitch URL paste (not own)	Metadata + embed preview only	Must be broadcaster/editor	Otherwise guide user to request editor rights
YouTube URL	Metadata via Data API; prompt to upload the original file (Studio download)	Channel ownership check via OAuth	No server download
Kick URL	Chat recorded live via official API, synced to the uploaded file (3.8)	Attestation	Re-evaluate if Kick publishes VOD API
Live stream	RTMP simulcast to our ingest (3.8), phase 3. Also Twitch "clip earlier in current stream" from chat spikes via EventSub	Broadcaster/editor	No live media ingest

Duplicate detection: SHA-256 of first+last 16 MB plus duration and size → warn and link to existing asset. Failed uploads resume from last part; sessions expire after 24 h.

3.4 Highlight detection pipeline

Scoring combines cheap signals over a sliding window, then an LLM judges only the top ~40 windows.

Signal	Source	Feasible	Cost
Transcript (word timestamps)	STT on extracted 16 kHz mono audio	Yes	$0.15–$0.46/h
Loudness / RMS spikes, pitch variance, speech rate	FFmpeg ebur128 + librosa	Yes	CPU, negligible
Laughter, cheering, screams	YAMNet/PANNs audio-event model	Yes, moderate accuracy	CPU
Scene/visual change	PySceneDetect on 2 fps proxy	Yes	CPU
Face presence/expression	MediaPipe on facecam crop	Feasible; privacy notice, opt-in for expression	CPU
Chat velocity, emote bursts	Twitch chat captured live via EventSub/IRC by our bot with broadcaster consent; no official historical chat API	Only if connected before the stream	Free
Game events (kills, wins)	OCR/templates per game	Hard; defer	—
Story completeness, hook, novelty	LLM over transcript windows	Yes	≈$0.01–$0.05 per VOD hour with a small model

Clip Quality & Engagement Potential Score (0–100). The UI states that the score measures hook strength, emotional energy and narrative completeness. It does not predict views, which depend on platform algorithms, posting time and audience size.

For a candidate window w = [t_start, t_end], five sub-scores are computed, each normalized to [0, 1]:

- **Hook strength (S_hook):** evaluates the first 3.5 s, [t_start, t_start + 3.5 s], for curiosity gaps, bold claims, immediate vocal intensity or visual shock. Scored by the LLM from the transcript plus audio and visual flags.
- **Emotional and acoustic intensity (S_energy):**

```
S_{energy}(w) = \sigma\left(\alpha_1 \max_{t \in w} Z_{RMS}(t) + \alpha_2
\max_{t \in w} Z_{F_0}(t) + \alpha_3 \, P_{laughter/scream}(w)\right)
```

σ is a calibrated sigmoid. Z_RMS and Z_F0 are loudness and pitch z-scores against the stream's own baseline (rolling 10-minute window, so loud streamers aren't always "exciting"). P is the audio-event model's probability.

- **Audience reaction velocity (S_chat):** computed when a chat log exists:

```
S_{chat}(w) = \min\left(1,\; \frac{\text{PeakMsgRate}(w) - \mu_{chat}}
{4\,\sigma_{chat}} + 0.25\cdot\mathbb{I}(\text{"clip it" in } w)\right)
```

Floor at 0. Peak rate uses a 5 s sliding window, shifted back 3–8 s to account for chat lag behind the on-stream moment. Without chat, w_c is redistributed to w_e and w_n in proportion to their weights.

- **Narrative and context completeness (S_narrative):** LLM rating (0–1) of whether a cold TikTok viewer understands the clip without stream context, and whether it has setup, climax and resolution.
- **Pacing and dead-air penalty (P_dead):** share of the clip's duration in silent stretches longer than 1.2 s (VAD = 0 and low visual motion).

Composite, configurable per creator niche preset:

```
C_{score}(w) = \text{clip}_{[0,100]}\left(100\left(w_h S_{hook} + w_e
S_{energy} + w_c S_{chat} + w_n S_{narrative} - w_p P_{dead}\right)\right)
```

The positive weights in each preset sum to 1.0, so a perfect clip with no dead air scores 100.

Niche preset	Hook w_h	Energy w_e	Chat w_c	Narrative w_n	Dead-air penalty w_p
GAMING (competitive FPS/MOBA)	0.20	0.35	0.25	0.20	0.20
JUST_CHATTING / reaction	0.30	0.25	0.20	0.25	0.25
IRL (provisional)	0.25	0.30	0.20	0.25	0.20
PODCAST / interview	0.35	0.15	0.05	0.45	0.30
EDUCATIONAL (provisional)	0.35	0.10	0.05	0.50	0.30

The IRL and EDUCATIONAL rows are starting guesses. All weights are defaults to tune against the hand-labeled set in the week-2 spike. Per-creator weights adapt only after 20 or more published clips (section 7.2). Novelty is not a weight; it is applied as a dedup filter (below).

Boundaries: start at the sentence start preceding the peak (search back ⩽ 20 s for a transcript sentence boundary, then snap to a silence ⩾ 150 ms); end after the reaction settles (audio energy returns to baseline + 1.5 s, ending on a word boundary). Pad 0.3 s; enforce min/max length; never cut inside a word.

Dedup: reject candidates overlapping > 40% in time or > 0.85 transcript-embedding similarity with a higher-scored candidate. Reject < 70% speech-or-event coverage, black/frozen frames > 30%, or length outside platform limits.

3.5 Clip generation and editing

Browser does: preview (HLS proxy at 720p), timeline trimming, caption text edits, layout selection, safe-zone overlays, undo/redo (command stack), autosave every 5 s. Backend does: all final renders, reframing analysis, waveform/thumbnail strips, transcription.

Edit state is a JSON **render spec** (versioned), so the preview and FFmpeg render read the same document:

```
{"version":3,"source":"ma_123","in":4521.30,"out":4558.90,"aspect":"9:16","la
yout":"split","regions":{"face":[1580,40,320,320],"game":
[0,0,1920,1080]},"captions":{"style":"bold_pop","position":0.72},"brand":
{"logo":"as_9","watermark":false},"audio":{"normalize":-14}}
```

Layouts at launch: centered crop with face-tracking, split-screen (facecam top / game bottom), letterboxed fit. Later: PiP, zoom punches, intros/outros, music library (licensed only, e.g. Epidemic/Artlist business license — verify terms).

Rendering: FFmpeg 7 with libx264 (CPU) at launch; NVENC GPU workers only when queue time breaches SLO. Normalize VFR to CFR (`-fps_mode cfr`), loudness to −14 LUFS, captions burned via libass (ASS file generated from word timings). Fonts bundled in the worker image to avoid missing-font failures. Remotion or browser-side WebCodecs considered and rejected for v1 (licensing cost/complexity, inconsistent device support).

3.6 Captioning

Option	Price	Strength	Weakness
AssemblyAI Universal-2 batch	≈$0.15/h	Cheap, diarization, 99 languages	Vendor dependency
Deepgram Nova-3 batch	≈$0.26/h (streaming ≈$0.46/h)	Keyterm prompting, speed	Slightly pricier
Whisper large-v3 self-hosted (faster-whisper)	GPU ≈$0.001–$0.002/min at rented rates	No vendor, MIT license	Ops burden; hallucination on silence/music

Prices from third-party summaries (convertaudiototext); confirm on vendor pages before contracting. Launch on AssemblyAI with Deepgram as failover behind one interface. Custom vocabulary from creator profile (names, game terms). Profanity: keep verbatim by default, optional mask in captions. Captions export as SRT/VTT plus burned-in.

3.7 Publishing

State machine per publish job: `draft → queued → uploading → processing → published | failed | needs_user_action` . Each job carries an idempotency key (clip_version_id + account_id + destination); before any retry the worker checks the platform for an existing post id to prevent duplicates. Scheduled posts are executed by a scheduler that enqueues 10 minutes early; Instagram containers created no earlier than 1 hour before publish (24 h expiry). Account shown with avatar + handle in the confirm dialog to prevent wrong-account posts.

3.8 Additions adopted from the second draft review

Five ideas from an alternative blueprint are adopted; its platform claims were checked against section 2 and not taken over where they conflict.

Addition	How it works	Phase	Cost / risk
RTMP simulcast ingest	Streamer adds a second output in OBS (or Restream) to `rtmp://ingest.{domain}/live/{per-user key}`. An ingest node (SRS or nginx-rtmp) segments to 10 s HLS chunks in R2, keeping a rolling 2-hour buffer; the stream becomes a media asset on `stream.offline`. Enables clipping while still live.	3 (beta)	Doubles streamer upload bandwidth; we pay ingest compute and storage per live hour (≈0.6–3 GB/h at 1.5–6 Mbps). Plan-gated, with max concurrent streams per plan. Our key is ours, never the streamer's Twitch/Kick stream key.
Kick chat sync	With Kick OAuth, subscribe to chat events during the stream and store messages with wall-clock timestamps. After upload, align by the recording's creation time, refined by matching our RTMP audio or the stream-start event; confirm with a user-adjustable offset. Feeds S_chat.	3	Alignment errors of a few seconds; covered by the 3–8 s chat-lag window.
"Request permission to clip" link	An editor pasting an unauthorized channel gets a signed invite link (expires in 14 days). The streamer opens it, signs in with Twitch/YouTube/Kick, picks permitted uses, and a `creator_grants` row is created.	MVP	Doubles as low-cost acquisition: every editor invite brings a streamer.

Addition	How it works	Phase	Cost / risk
Coarse-to-fine triage	Stage 1 runs on the whole recording on CPU: loudness, voice activity (Silero VAD), audio events, chat velocity. It keeps the most active ~20–30% of windows. Stage 2 transcribes the full audio (cheap, needed for search) but runs the LLM only on surviving windows. Stage 3 runs face tracking and boundary snapping only on candidates.	MVP	Savings mainly on LLM and vision calls; real percentage to be measured, not assumed.
Stream-friendly metering	Long VODs are billed at a discounted triage rate (e.g. 0.25 credit per hour for dead-air stretches detected by Stage 1) plus full rate for analyzed windows. Removes "credit burn" on 6-hour streams versus per-source-minute competitors.	4	Must be simple to explain; test against flat source-hour pricing in the price test.

Also adopted: lossless 30-minute segmentation of long recordings (`ffmpeg -f segment -segment_time 1800 -c copy`) so analysis runs in parallel; camera smoothing with a deadband (crop moves only when the face leaves the inner 25% of the frame) plus easing; automatic layout switching when the streamer changes OBS scenes (detected by scene-change + face-box jumps); optimistic locking on clip edits (`409 version_conflict`); TikTok "upload to inbox" as the default when the creator wants to add trending sounds.

Not adopted: asking for Twitch `channel:read:stream_key` (security risk, not needed); Remotion for rendering (paid company license, browser-render variance); music removal (Demucs) on every render, kept as an opt-in GPU feature in phase 5; the unsourced "2-hour stream in 2 minutes" benchmark.

4. Architecture, API, database and jobs

Chosen: a **modular monolith** — one FastAPI API process plus specialized worker pools sharing one codebase, Postgres, Redis and S3-compatible storage. Split services only when a module's scaling or deploy cadence demands it.

4.1 Options compared

Option	Pros	Cons	Verdict
A. Modular monolith: Next.js + FastAPI + Celery/Redis (or Dramatiq) + FFmpeg workers on Fly.io/Render/Hetzner + Cloudflare R2	Python owns AI/media ecosystem; one repo; cheap; R2 has zero egress fees	Two languages (TS + Python)	**Selected**
B. Full TypeScript: Next.js + Node + BullMQ + FFmpeg via child processes	One language	Weaker ML ecosystem; Python still needed for audio models	Viable alternative
C. Serverless (Lambda/Cloud Run jobs + Step Functions)	Scales to zero	15-min/size limits fight 2-hour media; cost opaque	Rejected for v1
D. Microservices on Kubernetes	Independent scaling	Huge ops cost for 1–4 devs	Rejected until > 50k MAU

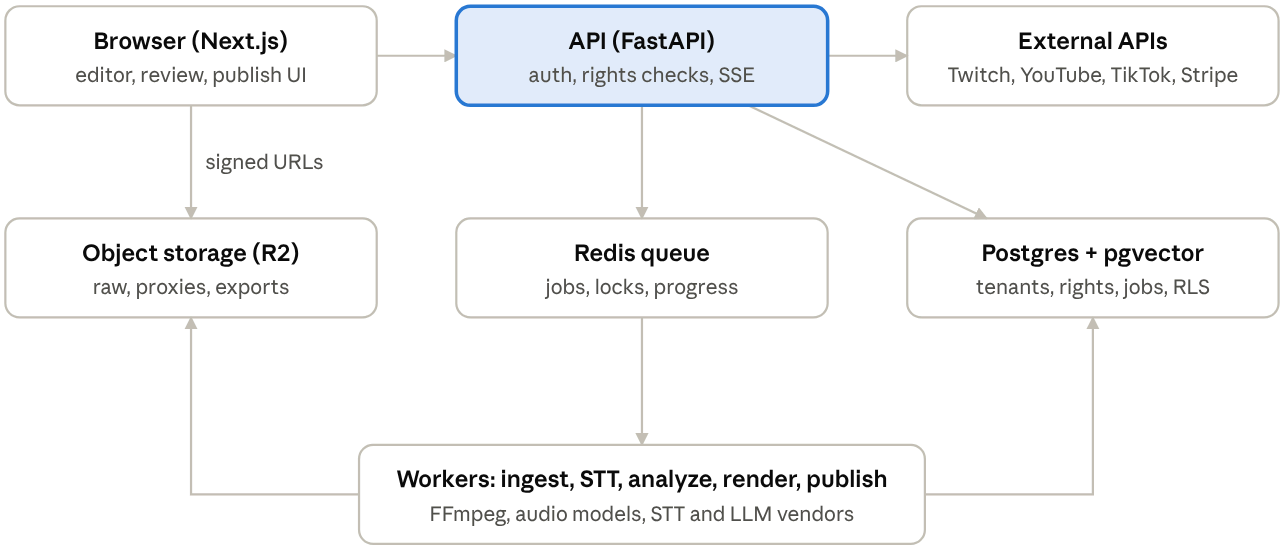
4.2 Components

Component	Choice	Why	Failure behaviour	Scales by
Web app	Next.js (App Router), TypeScript, Tailwind, shadcn/ui; Vercel or same host	Fast UI iteration	Static marketing still served	CDN

Component	Choice	Why	Failure behaviour	Scales by
API	FastAPI + Pydantic v2 + SQLAlchemy 2	Typed contracts, OpenAPI generation for TS client	5xx → frontend retries idempotent GETs	Horizontal replicas
Queue	Redis 7 + Celery (queues: `ingest`, `stt`, `analyze`, `render`, `publish`, `default`)	Mature retries, priorities	Jobs persist in Postgres `jobs` table; reconciler re-enqueues	More workers per queue
DB	Postgres 16 (Neon/Supabase/RDS) + pgvector	Relational + embeddings in one place	Read-only mode banner; workers pause	Vertical, then read replica
Object storage	Cloudflare R2 (S3 API)	No egress fees for downloads/previews	Uploads fail fast with resumable retry	Unlimited
Media workers	Docker image: FFmpeg 7, PySceneDetect, MediaPipe, YAMNet	Heavy CPU	Visibility timeout → job retried on another worker	Autoscale on queue depth
STT	AssemblyAI → Deepgram failover	Cost/quality	Job waits, user notified	Vendor
LLM	Small/medium hosted model via one adapter (e.g. Claude Haiku-class, GPT-mini-class)	Hook/context scoring, titles	Falls back to heuristic score	Vendor

Component	Choice	Why	Failure behaviour	Scales by
Realtime progress	Server-sent events from API, fed by Redis pub/sub	Simpler than WebSockets	Client polls every 5 s	—
Auth	Own (FastAPI + sessions) or Clerk/Supabase Auth	—	—	—
Billing	Stripe Billing + Paystack for NGN cards	Global + Nigeria	Grace period, no data loss	—
Observability	Sentry, Grafana Cloud (Prometheus/Loki), Better Stack uptime	—	—	—

One API and one worker pool share Postgres, Redis and object storage



MVP architecture · modular monolith

The browser uploads directly to storage with signed URLs; the API only enqueues work, and workers write results to Postgres and storage.

4.3 Recording lifecycle

1. **Authorize**: rights attestation or verified Twitch role → `content_rights` row.
2. **Ingest**: resumable upload to `raw/{workspace}/{asset}`; checksum; `ffprobe` validation.
3. **Prepare**: transcode 720p HLS proxy + 16 kHz mono WAV + 2 fps frame proxy (one pass, `ffmpeg` multi-output).
4. **Transcribe**: STT with word timestamps → `transcript_segments`; embeddings (per 30 s chunk) → pgvector.
5. **Analyze**: audio features, events, scene changes, chat → windows; heuristic score; LLM rerank top 40 → `highlight_candidates`.
6. **Edit**: user opens candidate → `clips` + `clip_versions` (render spec).
7. **Render**: worker reads source range with fast seek (`-ss` before `-i`, then accurate trim) → MP4 to `exports/`.
8. **Deliver**: signed download URL (15 min) or publish job.
9. **Measure**: analytics pull jobs at +1 h, +24 h, +7 d, +28 d where APIs allow.
10. **Retain/delete**: raw source deleted after 30 days (Free 7) unless pinned; proxies 90 days; exports until user deletes; hard-delete within 30 days of account closure.

4.4 API (v1, user-facing)

All routes `/api/v1`, JSON, cookie session (web) or bearer token (API plan). Errors use RFC 9457 problem+json: `{"type","title","status","detail","code","request_id"}`. Every POST that creates work accepts `Idempotency-Key` (stored 24 h). Authorization: every resource resolves to a `workspace_id`; membership role checked in a single dependency.

Method	Route	Purpose	Role	Rate limit
POST	/auth/signup, /auth/login, /auth/logout	Session auth	public	10/min/IP
POST	/auth/verify-email, /auth/password/reset, /auth/2fa/enable	Account security	public/user	5/min
GET/PATCH	/me	Profile, preferences	user	60/min

Method	Route	Purpose	Role	Rate limit
GET/POST	/workspaces; /workspaces/{id}/members	Workspaces, invites	owner/admin	30/min
GET	/connections/{provider}/authorize	Start OAuth (state + PKCE)	member	10/min
GET	/connections/{provider}/callback	Finish OAuth	—	—
DELETE	/connections/{id}	Revoke + delete tokens	admin	—
POST	/uploads	Create resumable upload session (returns part URLs)	editor+	20/h
POST	/uploads/{id}/complete	Finalize, create media asset	editor+	—
POST	/imports	Import from Twitch clip / Drive (`source`, `ref`, `rights`)	editor+	20/h
GET	/media, /media/{id}	Library, metadata, status	viewer+	—
GET	/media/{id}/transcript	Segments with word timings	viewer+	—
GET	/media/{id}/candidates?sort=score	Highlight candidates	viewer+	—
POST	/media/{id}/search	NL search (phase 5)	viewer+	30/min
POST	/clips	Create clip from candidate or manual range	editor+	—
GET/PATCH	/clips/{id}	Read/update (PATCH creates a new version)	editor+	—
GET	/clips/{id}/versions	Version history	viewer+	—
POST	/clips/{id}/renders	Queue render (preset) — idempotent	editor+	per plan concurrency

Method	Route	Purpose	Role	Rate limit
GET	/renders/{id}	Status, progress, download URL	viewer+	—
GET	/jobs/{id}/events	SSE progress stream	viewer+	—
DELETE	/jobs/{id}	Cancel	editor+	—
POST	/publications	Publish/schedule (`render_id`, `connection_id`, `caption`, `privacy`, `scheduled_at`) — idempotent	publisher+	—
GET/DELETE	/publications/{id}	Status / cancel scheduled	publisher+	—
GET	/analytics/clips?from=&to=	Performance snapshots	viewer+	—
GET	/usage	Quota consumption	member	—
POST	/billing/checkout, /billing/portal	Stripe/Paystack sessions	owner	—
POST	/webhooks/stripe, /webhooks/paystack, /webhooks/twitch	Signed webhooks	signature	—
POST	/me/export; DELETE /me	GDPR/NDPA export and deletion	user	1/day
GET/POST	/admin/*	Users, abuse reports, takedowns, refunds	staff (separate SSO)	—

Example:

```
POST /api/v1/clips/cl_8f2/renders
Idempotency-Key: 3b1c…
{"version_id":"cv_14","preset":"vertical_1080p"}

202 Accepted
{"render_id":"rd_77","status":"queued","credits_reserved":1,"events_url":"/api/v1/jobs/jb_91/events"}
```

Errors: `402 quota_exceeded`, `409 render_in_progress`, `422 invalid_range` (out outside media duration, length outside plan limits), `403 rights_missing`.

Internal interfaces are Python protocols, not HTTP: `SourceProvider`, `TranscriptionProvider`, `LLMProvider`, `Publisher`, `Storage`, each with a fake for tests.

4.5 Database schema

Core tables (all with `id uuid pk`, `created_at`, `updated_at`; tenant tables carry `workspace_id` and Postgres row-level security keyed on `current_setting('app.workspace_id')`):

Table	Key fields	Notes
users	email (citext unique), password_hash, email_verified_at, totp_secret_enc, deleted_at	
workspaces / memberships	plan_id; role enum(owner, admin, editor, publisher, viewer, client_reviewer)	unique(workspace_id,user_id)
connections	provider, external_account_id, scopes[], access_token_enc, refresh_token_enc, expires_at, status	AES-GCM envelope encryption (KMS key)
content_rights	media_asset_id, grantor_type(owner/creator_grant/license/twitch_role), grantor_user_id, permitted_uses[] (edit, download, publish, commercial, white_label, monetize), expires_at, revoked_at, evidence_ref	Checked on render/publish
creator_grants	creator_user_id, grantee_workspace_id, uses[], channels[], expires_at	One-click editor authorization
media_assets	source_type, source_ref, sha256, duration_s, width, height, fps, codecs, storage_keys jsonb, status, retain_until	index(workspace_id, created_at desc); unique(workspace_id, sha256) partial where not deleted
transcripts / transcript_segments	provider, language; segments: start_ms, end_ms, speaker, text, words jsonb, embedding vector(1024)	ivfflat/hnsw index on embedding
signals	media_asset_id, kind, t_ms, value	Bulk-insert via COPY; partition by asset if large

Table	Key fields	Notes
highlight_candidates	start_ms, end_ms, score, components jsonb, title, reason, status	index(media_asset_id, score desc)
clips / clip_versions	clip: candidate_id, title; version: spec jsonb, number, created_by	Immutable versions
render_jobs / exports	clip_version_id, preset, status, progress, error_code, storage_key, bytes	unique(clip_version_id, preset) where status != failed
publications	export_id, connection_id, destination, idempotency_key unique, status, scheduled_at, external_post_id, url, error	
analytics_snapshots	publication_id, captured_at, views, likes, comments, shares, avg_watch_s, raw jsonb	
plans / subscriptions / usage_records / payments / refunds	provider ids, period, credits_used, meter	usage_records append-only
webhook_events	provider, event_id unique, payload, processed_at	Dedup
jobs	type, status, attempts, run_after, lease_until, payload, idempotency_key unique	Source of truth for queue
audit_logs	actor_id, action, target, ip, metadata	Append-only, 1 year
abuse_reports / takedowns	reporter, target, reason, status, resolution	

Representative migration:

```
CREATE TABLE content_rights (
  id uuid PRIMARY KEY DEFAULT gen_random_uuid(),
  workspace_id uuid NOT NULL REFERENCES workspaces(id) ON DELETE CASCADE,
  media_asset_id uuid NOT NULL REFERENCES media_assets(id) ON DELETE CASCADE,
  grantor_type text NOT NULL CHECK (grantor_type IN
('owner','creator_grant','license','twitch_role')),
  grantor_user_id uuid REFERENCES users(id),
  permitted_uses text[] NOT NULL,
  attestation_text_version text NOT NULL,
  attested_ip inet,
  expires_at timestamptz,
  revoked_at timestamptz,
  created_at timestamptz NOT NULL DEFAULT now()
);
CREATE INDEX ON content_rights (media_asset_id) WHERE revoked_at IS NULL;
ALTER TABLE content_rights ENABLE ROW LEVEL SECURITY;
CREATE POLICY tenant ON content_rights USING (workspace_id =
current_setting('app.workspace_id')::uuid);
```

Account closure: users anonymized (email hashed), tokens deleted and revoked at provider, media and exports hard-deleted, payments/invoices retained 7 years (tax), audit logs retained 1 year with actor pseudonymized.

4.6 Background jobs

States: `queued → running → succeeded | failed | cancelled`, with `running → retrying → running` and `failed → dead` after max attempts. Rules:

- Retries: exponential backoff with jitter (30 s, 2 m, 8 m, 30 m); max 4; non-retryable codes (corrupt file, rights revoked) fail immediately.
- Leases: worker heartbeats every 30 s; reconciler re-queues jobs whose lease expired (crash recovery). Every step writes outputs to deterministic keys so reruns overwrite, not duplicate.
- Priorities: paid queue weight 3, free weight 1; per-workspace concurrency (Free 1, Creator 2, Pro 4, Agency 10) enforced with Redis semaphores.
- Timeouts: STT 60 min, render 15 min per clip, ingest 2 h. Temp files in per-job dir, deleted in `finally`; nightly sweeper.
- Dead-letter: `dead` jobs alert Slack and appear in admin console with one-click retry.

Worked example: 2-hour VOD → 10 candidates (assumptions: 1080p60 H.264, 6 GB, 8-vCPU worker):

Step	Wall time	Resource
Upload (user, 50 Mbps)	≈16 min	user bandwidth
Proxy + audio extract	6–10 min	8 vCPU
STT (batch API, parallel chunks)	3–8 min	vendor; ≈$0.30–$0.52
Audio/visual signals	4–6 min	4 vCPU
LLM rerank of 40 windows	1–2 min	≈$0.02–$0.10
Candidates ready after upload	**≈12–20 min**	
Render 10 × 45 s vertical clips	≈1–2 min each, 4 in parallel → 4–5 min	

Targets to validate in the week-2 spike; these are estimates.

5. External services and credentials

The MVP needs about 15 secrets; social publishing adds credentials that only become useful after each platform's review. Nothing below grants access by itself — every provider requires an account, app registration and, for publishing, approval.

5.1 Service inventory

Service	Purpose	Auth	Approval needed	Free / price (verify)	Failure impact → fallback
Twitch Helix	OAuth login, VOD metadata, Create Clip/From VOD, clip download, EventSub chat	OAuth2 (client id/secret), app + user tokens	None for basic; Clip-from-VOD is open beta	Free; per-token rate-limit bucket (Ratelimit-* headers)	Twitch import disabled → upload
Kick public API	Login, channel metadata, chat events	OAuth 2.1 + PKCE	Developer app	Free	Metadata only
YouTube Data API v3	Channel verify, metadata, Shorts upload	OAuth (`youtube.upload` , `youtube.readonly`) + API key for public reads	Google OAuth verification + YouTube API compliance audit for public uploads	10,000 units/day default	Publish → download fallback
TikTok Content Posting API	Direct post / inbox upload	OAuth (`user.info.basic` , `video.publish` , `video.upload`)	App review + content audit for public posts; domain verification for pull_from_url	Free	Private-only until audited
Instagram Graph API	Reels publish, insights	Instagram Login (`instagram_business_content_publish`) or Facebook Login (`instagram_content_publish`)	Meta App Review + Business Verification	Free	Phase 4

Service	Purpose	Auth	Approval needed	Free / price (verify)	Failure impact → fallback
X API v2	Post + media	OAuth 2.0 PKCE (user)	Developer account, credits	≈$0.015/post	Phase 4, pass cost through
AssemblyAI / Deepgram	STT	API key	No	≈$0.15–$0.46/h	Failover to the other
LLM provider	Hook/context scoring, titles, captions	API key	No	Per token	Heuristic-only scoring
Embeddings	Search, dedup	API key or self-hosted (bge-m3)	No	Small	Text-only dedup
Cloudflare R2	Media storage	Access key + secret	No	$0.015/GB-month, zero egress (verify)	Uploads blocked
Postgres (Neon/Supabase)	DB	Connection string	No	Free tier → ≈$19–$69/mo	Read-only mode
Redis (Upstash/managed)	Queue, locks	URL	No	≈$10–$30/mo	Jobs held in Postgres
Stripe Billing	Global subscriptions	Secret + webhook secret	Account activation	2.9% + 30¢ US cards (+intl fees)	Grace period
Paystack	NGN cards, bank transfer	Secret key	Business verification	Local % fee (verify)	Stripe
Resend / Postmark	Email	API key	Domain DNS	Free tier	Queue and retry
Sentry, Grafana Cloud, Better Stack, PostHog	Errors, metrics/logs, uptime, product analytics	DSN / keys	No	Free tiers	Degraded visibility
GitHub Actions	CI/CD	OIDC to host	No	Free minutes	Manual deploy

5.2 .env.example

```
# --- MVP required (private, server-only) ---
DATABASE_URL=postgresql://user:pass@host:5432/clipforge
REDIS_URL=rediss://default:pass@host:6379
SESSION_SECRET=                 # 64 random bytes, base64
TOKEN_ENCRYPTION_KEY_ID=        # KMS key reference for envelope encryption
```

```
R2_ACCOUNT_ID=
R2_ACCESS_KEY_ID=
R2_SECRET_ACCESS_KEY=
R2_BUCKET_MEDIA=clipforge-media
ASSEMBLYAI_API_KEY=
LLM_API_KEY=
STRIPE_SECRET_KEY=
STRIPE_WEBHOOK_SECRET=
RESEND_API_KEY=
SENTRY_DSN=

# --- MVP integrations ---
TWITCH_CLIENT_ID=               # public identifier
TWITCH_CLIENT_SECRET=
TWITCH_EVENTSUB_SECRET=         # HMAC secret for webhook verification
GOOGLE_CLIENT_ID=               # YouTube OAuth (public identifier)
GOOGLE_CLIENT_SECRET=
TIKTOK_CLIENT_KEY=              # public identifier
TIKTOK_CLIENT_SECRET=

# --- Optional / failover ---
DEEPGRAM_API_KEY=
PAYSTACK_SECRET_KEY=
POSTHOG_PROJECT_KEY=            # may be exposed to browser

# --- Only after platform approval (phase 4) ---
META_APP_ID=
META_APP_SECRET=
X_CLIENT_ID=
X_CLIENT_SECRET=
KICK_CLIENT_ID=
KICK_CLIENT_SECRET=

# --- Not needed if self-hosting Whisper / embeddings ---
# WHISPER_ENDPOINT=http://stt-gpu:8000

# --- Browser-safe (NEXT_PUBLIC_ prefix only) ---
NEXT_PUBLIC_API_URL=https://api.example.com
NEXT_PUBLIC_STRIPE_PUBLISHABLE_KEY=
```

5.3 Secret handling

- Store in the host's secret manager (Fly secrets / Doppler / AWS Secrets Manager); never in the repo. CI uses OIDC, not long-lived keys.
- OAuth redirect URIs registered per environment: `https://app.{domain}/api/v1/connections/{provider}/callback`, plus `http://localhost:3000/...` in a separate dev app.
- User tokens encrypted with a per-row data key wrapped by a KMS master key; rotation = rewrap job. Refresh tokens rotated on every refresh where the provider supports it.
- Rotation: provider secrets every 90 days or on staff departure; dual-key acceptance window for webhook secrets.
- Revocation: disconnect calls the provider revoke endpoint, then deletes ciphertext.

6. Compliance, security and failure handling

Rights are enforced in code: no render, download or publish happens without an unrevoked `content_rights` record covering that use.

6.1 Content rights model

Use level	Who can grant	Default for owner upload	Default for editor via creator grant
Personal edit / internal review	Owner, creator grant	Yes	Yes
Download	Owner, creator grant	Yes	Per grant
Social publishing (own accounts)	Owner	Yes	Only to creator's connected accounts unless grant says otherwise
Commercial redistribution / monetized	Owner, written license	Attest separately	Per grant
White-label client work	License	No	Per grant

Revocation or expiry triggers: block new renders/publishes, cancel scheduled posts, notify workspace, keep existing exports downloadable only by the owner for 7 days, then delete. Published posts on third-party platforms cannot be recalled by us; the UI lists them so the user can remove them.

Legal position (requires counsel review before launch, especially for US, EU/UK and Nigeria):

- Public availability is not permission. Fair use is not assumed.
- Register a DMCA agent (US Copyright Office) and publish a takedown/counter-notice process; repeat-infringer policy (3 strikes → termination).
- Music: stream VODs often contain copyrighted music. Warn when audio-fingerprint or music-detection confidence is high; offer "mute music" (source separation via Demucs, phase 5). We do not clear music rights.
- Privacy: GDPR/UK GDPR, Nigeria Data Protection Act 2023, CCPA. Voice and face data processed only for the user's own editing; no biometric identification, no cross-customer training without opt-in. DPA with every subprocessor.
- Terms: platform developer agreements (Twitch, Google API Services, TikTok, Meta) require displaying their branding/permissions accurately and honoring user deletion requests.

6.2 Security controls

- **Tenant isolation:** workspace_id on every row + Postgres RLS + API dependency check; object keys prefixed by workspace and only reachable via short-lived signed URLs (15 min download, 1 h upload part). Automated test tries cross-tenant access on every endpoint.
- **OAuth:** state + PKCE, exact redirect matching, tokens encrypted, refresh-token rotation, least scopes.
- **Web:** SameSite=Lax HttpOnly session cookies + CSRF token on mutations; strict CSP; React escaping; caption text sanitized before ASS generation (escape `{}` override tags).
- **Injection:** SQLAlchemy parameterized queries only; FFmpeg invoked with argv arrays, never shell strings.
- **SSRF:** server-side fetches limited to allowlisted provider hosts; block private IP ranges; no arbitrary URL fetch.

- **Uploads:** ffprobe validation in a sandboxed container (no network, seccomp, CPU/mem/time limits); reject > plan size; cap resolution at 4K and duration at plan max; ClamAV scan for non-video types.
- **Abuse:** rate limits per IP and account, Turnstile on signup, email verification before processing, disposable-email blocklist, free-tier abuse heuristics (same card/device across accounts).
- **Billing:** entitlements derived only from verified webhooks; webhook signature check + event-id dedup; usage reserved before a job and settled after.
- **Audit log:** logins, connection changes, rights grants, publishes, deletions, admin actions.
- **Incident response:** severity levels, on-call rota, 72-hour regulator notification path, token mass-revocation runbook.

6.3 Edge-case and failure matrix

Case	Detection	User sees	Backend action	Billing impact
Invalid / unsupported URL	Regex + provider lookup	"We can't import from this link. Upload the file instead."	No job	None
Private, deleted or region-locked source	Provider 403/404	"This video isn't available to your account."	Mark failed, non-retryable	None
Stream still live / VOD not ready	Twitch stream status	"We'll import when the VOD is ready."	Poll every 10 min for 6 h	None
Interrupted / expired upload	Missing parts / session > 24 h	Resume button	Keep parts 24 h, then abort multipart	None
Corrupt, zero-duration, no audio/video	ffprobe	Specific message (e.g. "No audio track — captions unavailable")	Fail or continue in video-only mode	Refund reserve
VFR, odd codec (HEVC 10-bit, AV1)	ffprobe	Nothing	Normalize in proxy pass	—

Case	Detection	User sees	Backend action	Billing impact
Duplicate upload	sha256	"Already in library" with link	No new processing	None
Transcription fails / unsupported language	Vendor error / low confidence	"Captions unavailable for this language"	Failover vendor; continue with audio/visual signals	Partial
No good highlights	Max score < threshold	"No strong moments found — try search or manual clip"	Show top 5 anyway, flagged low	Charged (processing happened) — consider half credit
LLM outage / cost spike	Error rate, $/hour alert	Slightly slower results	Heuristic scoring; daily spend cap per workspace and global	—
Hallucinated titles	LLM output must cite transcript span	Editable suggestion	Reject outputs not grounded in span	—
Worker crash / OOM / disk full	Lease expiry, exit code	Progress continues after retry	Retry on larger instance class; sweeper frees disk	Reserve held
Missing font, broken subtitles	Render validation (frame sample + ASS parse)	"Render failed, retrying"	Fallback font	Refund on final fail
A/V desync	Compare stream durations > 100 ms	—	Re-render with `aresample=async=1`	—
User cancels render	Cancel flag checked per segment	Cancelled	Kill ffmpeg, delete temp	Refund
Token expired / revoked	401 from provider	"Reconnect TikTok to publish" (email + in-app)	Pause job `needs_user_action`	—
Platform rate limit	429 / quota error	"Scheduled for HH:MM"	Reschedule after reset	—
Duplicate publish risk	Idempotency key; pre-retry status check	—	Never retry a non-idempotent upload blindly	—

Case	Detection	User sees	Backend action	Billing impact
Scheduled post misses window	Scheduler lag > 5 min	"Posted 12 min late" or "Failed — post now?"	Alert ops	—
Post removed by platform	Analytics fetch 404	Badge "Removed by platform"	Stop polling	—
App audit revoked	Provider error codes	Publishing to X paused banner	Feature-flag destination off	—
Payment failure	Stripe invoice.payment_failed	Banner, 7-day grace	Dunning emails; then downgrade to Free, keep data 30 days	—
Duplicate webhook	event_id unique	—	Ignore	—
Chargeback	dispute.created	Account flagged	Freeze processing, evidence pack	Fee absorbed
Downgrade over limits	Plan change	"3 of 5 seats active — choose which remain"	Read-only for excess	Prorated
Owner leaves workspace	Membership delete	Must transfer ownership first	Block	—
Account deletion / export	User request	Confirmation + 30-day window	Export zip within 72 h; hard delete pipeline	Cancel subscription

Severity: S1 (data leak, mass publishing failure, billing errors) pages on-call immediately; S2 (one integration down) within 1 h; S3 next business day.

7. UX, differentiating features, testing and operations

The UX target: a first-time streamer exports a usable vertical clip within 20 minutes of uploading, with no more than 6 clicks after candidates appear.

7.1 Navigation and screens

Left rail (desktop) / bottom tabs (mobile): **Home · Library · Clips · Publish · Analytics · Team · Settings**. Workspace switcher at top.

Screen	Key components	Empty / loading / error	Acceptance criteria
Home	Recent VODs with status, "Import" CTA, usage meter	Empty: upload dropzone + Twitch connect	Upload starts in ⩽ 2 clicks
Import	Dropzone, URL field with source detection, rights checkbox	Per-file progress with resume	Rights attestation required before Start
Processing	Step list (upload, prepare, transcribe, analyze) with ETA via SSE	Failure shows reason + action	Updates within 2 s of server event
Highlights	Candidate cards ranked: thumbnail, score bar with reason ("crowd-laugh + chat spike"), transcript snippet, length; keyboard J/K navigation, A=accept, X=reject	Low-confidence badge	Preview starts < 1.5 s on broadband
Editor	Player (left), transcript-as-timeline (click word to trim), layout picker, caption style panel, platform safe-zone toggle, undo/redo, version menu	Autosave indicator	Trim precision 1 frame; undo 50 steps
Export / Publish	Presets, destination accounts with avatars, caption + hashtag fields per platform, schedule picker, private/public	Platform limitation notes inline ("TikTok: private until our app is audited")	Confirm dialog names account
Calendar	Week/month view, drag to reschedule	—	Shows failures in red with retry
Analytics	Per-clip metrics, cohort by hook/length/layout	"Connect an account to see stats"	Labels correlations, not causes
Team	Members, roles, creator grants, approval queue	—	Role changes audited

Screen	Key components	Empty / loading / error	Acceptance criteria
Billing	Plan, usage, invoices, currency (USD/NGN)	—	Downgrade preview
Admin	User search, jobs/DLQ, takedowns, refunds, feature flags	—	Staff SSO + 2FA only

Visual direction: dark-first neutral UI (editors work at night next to OBS), one accent colour, Inter/Geist type, dense lists, no decorative gradients. WCAG 2.2 AA: captions editor fully keyboard-operable, focus rings, 4.5:1 contrast, reduced-motion respected. Mobile: review candidates, approve, schedule, download — no timeline editing in v1.

7.2 Differentiating features (ranked)

Rank	Feature	Value	Effort	Cost	Phase
1	Chat-aware highlight detection (Twitch bot records chat with consent)	High, unique vs podcast tools	M	Low	MVP
2	Creator → editor authorization (one-click grant, scoped uses)	High for editors, trust	M	Low	MVP
3	Transcript-as-timeline editing	Fast trimming	M	Low	MVP
4	Stream layout presets (facecam + gameplay split auto-detected)	High for gaming	M	Low	MVP
5	"Why this clip" explanations on every candidate	Trust, faster triage	S	Low	MVP
6	Natural-language search across VODs ("every Arsenal mention")	High	M	Embeddings	5
7	Per-creator ranking from performance feedback	Retention, moat	L	Low	5
8	Multi-clip campaign planner (10 clips → spaced calendar, variety rules)	Saves planning time	M	Low	5
9	Approval workflow + client reviewer role	Agencies pay	M	Low	6
10	Auto clip from current stream via Twitch "clip earlier in stream" on chat spikes	Wow factor	M	Low	5
11	Hook rewriter (first 2 s text overlay suggestions)	Quality	S	LLM	5

Rank	Feature	Value	Effort	Cost	Phase
12	Repost recycler: resurface past top clips for other platforms	Content volume	S	Low	5
13	Brand kits per creator (fonts, colours, logos)	Agencies	S	Low	4
14	Sponsor-segment finder ("find all sponsor reads") for sponsor reports	Helps creators earn	S	Low	5
15	Clip bounty / revenue-share ledger for creator-run clipping programs	Monetization, community	L	Low	6
16	Music detector + mute-music render	Avoids strikes	M	GPU	5
17	Multi-language caption translation	Reach	S	LLM	4
18	Weekly "best of stream" compilation auto-builder	YouTube long-form	M	Render	5
19	Caption template marketplace	Revenue	M	Low	7
20	Public API for agencies	Enterprise	M	Low	7

Feedback loop rule: show "clips under 30 s with a question hook got 1.8× median views for you (n=24)" with sample size and confidence; never claim causation. Personal weights update only after ≥ 20 published clips.

7.3 Testing

Layer	Tooling	Must cover
Unit	pytest, Vitest	Scoring, boundary snapping, dedup, render-spec → FFmpeg args, entitlements
Integration	pytest + Testcontainers (Postgres, Redis, MinIO)	Job state machine, retries, RLS isolation
Contract	OpenAPI schema diff in CI; generated TS client compiles	No breaking API changes
Media golden tests	20-file corpus (VFR, HEVC, mono, no audio, 4K, 6 h, corrupt)	Output duration ± 1 frame, A/V drift < 40 ms, loudness −14 ± 1 LUFS
Highlight quality	30 hand-labeled VOD hours	Precision@10 ≥ 0.6 before beta; tracked per release

Layer	Tooling	Must cover
Provider fakes + sandbox	Recorded fixtures; Stripe test mode; TikTok/YouTube sandbox accounts	OAuth expiry, 429, revoked scopes
E2E	Playwright	Signup → upload → candidate → edit → render → download → publish (fake) → billing usage recorded
Security	ZAP baseline, Semgrep, dependency audit, cross-tenant test suite, pentest before public launch	
Load	k6 for API; synthetic queue flood (500 VOD-hours)	p95 API < 300 ms; no starvation of paid queue
Accessibility	axe-core in Playwright	Zero critical violations
DR	Quarterly restore from backup to staging	RPO ⩽ 15 min, RTO ⩽ 4 h

MVP launch gate: all E2E happy paths green, golden media tests green, cross-tenant suite green, Precision@10 ⩾ 0.6, pentest criticals fixed, billing reconciliation matches Stripe for test month.

7.4 Deployment and operations

- **Environments:** local (Docker Compose: postgres, redis, minio, api, worker, web), staging (production clone with test keys), production.
- **MVP deploy:** web on Vercel; API + workers on Fly.io or Hetzner Cloud (CPU workers, autoscaled by queue depth via a small controller); Neon Postgres with PITR; R2; Upstash Redis. Move to Kubernetes only when > ~30 worker nodes or multi-region is needed.
- **CI/CD:** GitHub Actions — lint, typecheck, tests, build images, Alembic migrations (expand/contract only), deploy staging on main, manual promote to prod; instant rollback to previous image.

- **Dashboards & alerts:** queue depth and age per queue (alert if oldest paid job > 10 min); job failure rate > 5%/15 min; publish failure rate per destination; STT/LLM spend per hour vs budget (alert at 150% of 7-day average); storage growth per day; API 5xx > 1%; p95 candidates-ready time.
- **Backups:** Postgres PITR 7 days + nightly logical dump to separate account; R2 lifecycle rules; raw media not backed up (user can re-upload) — stated in terms.
- **Platform API changes:** subscribe to Twitch, Google, TikTok, Meta, X developer changelogs; contract tests against live sandboxes nightly.

8. Pricing, unit economics, financials and go-to-market

Meter on **source hours processed** (the true cost driver), sell on outcomes, and keep gross margin above 65% at typical usage. All prices are provisional until the validation price test.

8.1 Unit cost per source hour (estimate)

Item	Assumption	Cost
STT	Batch API, $0.15–$0.26/h	$0.15–$0.26
Analysis + proxy compute	≈2 vCPU-hours at $0.03–$0.05	$0.06–$0.10
LLM rerank + titles	≈40 windows, small model	$0.02–$0.10
Embeddings	120 chunks	< $0.01
Storage	3 GB raw × 30 days + 0.8 GB proxy × 90 days on R2	≈$0.08
Total per source hour		**≈$0.32–$0.55**
Render per 45 s clip (1080p, CPU)	≈0.2 vCPU-h	≈$0.01
Downloads / previews	R2 zero egress (verify plan terms)	≈$0
Publishing	Free except X ≈$0.015–$0.20/post	pass-through

Reuse: one transcript, proxy and embedding set serve all candidates, re-renders and searches; re-renders never re-transcribe. So 10 clips from 1 hour ≈ $0.45–$0.65; 100 clips/month (~10 h) ≈ $4.5–$6.5; 1,000 clips/month (~100 h) ≈ $45–$65; heavy agency (300 h, 3,000 clips) ≈ $130–$200.

Largest cost drivers: STT, free-tier usage, raw-file retention, and any GPU feature (music separation, self-hosted vision). Self-host Whisper only above ≈ 3,000 source hours/month, when a dedicated GPU beats API pricing.

8.2 Tiers

	Free	Creator	**Pro**	Agency	Enterprise
Monthly	$0	$15	**$29**	$149	Custom (from $600)
Annual (per month)	—	$12	$24	$124	Contract
Source hours / month	0.5	10	30	100	Committed
Max recording length	1 h	4 h	8 h	12 h	12 h+
Seats / creator workspaces	1 / 1	1 / 1	3 / 3	10 / 15	Custom
Concurrent jobs	1	2	4	10	Dedicated
Export	720p, watermark	1080p	1080p, 60 fps	4K	4K
Highlight AI	Basic	Full	Full + chat-aware	Full + chat-aware	+ custom models
Publishing / scheduling	Download only	3 accounts, publish	10 accounts, schedule	50 accounts, calendar	Unlimited
Analytics, brand kit	—	—	Yes	Yes + client reports	Yes
Approvals, white-label	—	—	—	Yes	Yes + SSO
Priority / support	Community	Email 48 h	Email 24 h, priority queue	Priority + chat	SLA, CSM
Overage per source hour	—	$1.50	$1.25	$1.00	Contract

Fair use: no automated bulk ingestion of third-party channels; publishing caps follow each platform's limits.

Nigeria and similar markets: offer local-currency pricing through Paystack at roughly 40–50% of USD list (purchasing-power adjusted), card and bank transfer, set after checking the exchange rate at launch. Restrict PPP pricing to local payment methods to limit arbitrage.

8.3 Margin by tier (estimate)

Tier	Typical use (40% of allowance)	Variable cost incl. payment fees	Gross margin	Heavy use (100%) margin
Creator $15	4 h, 40 clips	≈$3.0	≈80%	≈55%
Pro $29	12 h, 120 clips	≈$8.0	≈72%	≈35%
Agency $149	40 h, 400 clips	≈$29	≈80%	≈57%
Free	0.5 h	≈$0.25	—	—

Free-tier burden: at 5% conversion, each paying user carries ~19 free users ≈ $4.75/month. Keep Free at 30 minutes, require email verification, expire unused raw files after 7 days.

8.4 12-month projection (estimates, not verified data)

Assumptions — base: 400 signups in month 1 growing 20%/month, 5% free→paid within 60 days, 6% monthly paid churn, blended ARPU $27, variable cost 25% of revenue, fixed tooling/infra $800/month rising to $2,500, marketing $1,000→$4,000/month, founders unpaid. Conservative: 10% growth, 3% conversion, 9% churn, ARPU $22. Optimistic: 30% growth, 7% conversion, 4% churn, ARPU $32.

Scenario	Paying users M6	Paying users M12	MRR M12	ARR run-rate M12	Month cash-flow positive (excl. salaries)
Conservative	≈45	≈120	≈$2.6k	≈$32k	Not within 12 months
Base	≈110	≈400	≈$10.8k	≈$130k	≈Month 8
Optimistic	≈260	≈1,200	≈$38k	≈$460k	≈Month 5

CAC target ⩽ $60 blended (creator-led and content marketing); LTV at $27 ARPU, 72% margin, 6% churn ≈ $325 → LTV:CAC ≈ 5. Sensitivity: if users consume 100% of allowance and STT doubles in price, base gross margin falls from ~72% to ~45%; mitigations are lower included hours on Pro, overage billing, and moving STT in-house.

8.5 Go-to-market

1. **Validation:** 20 interviews (streamers 1k–50k followers, freelance clip editors), concierge pilot, landing page with $19/$29 price test.
2. **Private beta (50 creators):** free Pro in exchange for weekly feedback and usage data.
3. **Public launch:** Product Hunt, Twitch/YouTube creator communities, TikTok showing before/after clips.
4. **First 100 paying:** affiliate program (30% recurring for 12 months) for clip editors who bring their streamers; creator partnerships with mid-size streamers.
5. **First 1,000:** SEO on "Twitch VOD to TikTok", templates gallery, referral credits (+2 source hours each side).
6. **Teams/agencies:** outbound to clipping agencies; white-label and client billing.
7. **Enterprise:** esports orgs, MCNs, media companies with owned archives.

We do not sell user data or run ads in the product.

9. Roadmap, handoff and launch

Build the riskiest pieces first: highlight quality and platform audits. The UI is the cheapest part to change.

9.1 Phases

Phase	Duration	Features	Approvals	Gate to exit	Run cost/month
1 Validation	2–3 weeks	Interviews, concierge clipping with scripts, price page	Submit TikTok + YouTube audits, Twitch app	⩾ 10 users say they'd pay ⩾ $19; Precision@10 ⩾ 0.5 on labeled set	< $100
2 MVP	6–8 weeks	Upload, transcript, candidates, editor, render, download, Stripe, Twitch OAuth	Audits in progress	E2E green; 2-h VOD → candidates < 20 min	$200–$400

Phase	Duration	Features	Approvals	Gate to exit	Run cost/month
3 Private beta	4–6 weeks	Twitch clip import, YouTube/TikTok publish (private until audit), chat bot	Audits granted	50 creators, 40% week-4 retention, cost/hour within estimate	$400–$800
4 Public launch	4 weeks	Plans, Paystack, onboarding, support desk, Instagram (if approved), translation	Meta review	Pentest clean; first 50 paying	$600–$1,200
5 Advanced AI	8 weeks	NL search, personalization, campaigns, live chat-spike clips, music detection	—	Precision@10 ≥ 0.7; publish rate per candidate up 20%	$1–3k
6 Teams/agencies	6 weeks	Approvals, client reviewer, white-label, client billing, revenue-share ledger	—	10 agencies paying	$2–5k
7 Scale	Ongoing	Self-hosted STT, GPU render pool, SSO, public API, multi-region	—	Gross margin ≥ 75%	Scales with revenue

9.2 First 30 days

- [ ] Day 1–3: register Twitch, Google Cloud, TikTok developer apps; draft privacy policy and terms; submit TikTok and YouTube audits as early as their forms allow.
- [ ] Day 1–7: repo scaffold, Docker Compose, CI, Postgres schema v1, upload flow to R2.
- [ ] Day 5–12: STT adapter, signal extraction, heuristic scorer, LLM rerank; label 30 VOD hours (10 creators × 3 h).
- [ ] Day 12–14: measure Precision@10; tune weights; decide go/no-go on gameplay-heavy content.
- [ ] Day 14–21: candidate review UI, editor with render spec, FFmpeg renderer, caption burn-in, download.
- [ ] Day 21–26: auth hardening, Stripe plans + usage metering, Twitch OAuth + clip import behind flag.
- [ ] Day 26–30: staging soak test, 10 beta creators onboarded, cost dashboard live.

9.3 Repository layout

```
clipforge/
  apps/web/            # Next.js app (TS, Tailwind)
  apps/api/            # FastAPI: routers/, deps/, schemas/
  core/                # domain modules (shared by api + workers)
    identity/ workspaces/ rights/ media/ transcripts/
    highlights/ clips/ render/ publishing/ analytics/ billing/
  workers/             # Celery tasks per queue, ffmpeg/ helpers
  providers/           # twitch/ youtube/ tiktok/ meta/ x/ stt/ llm/ storage/
  migrations/          # Alembic
  packages/api-client/ # generated TS client from OpenAPI
  infra/               # docker-compose.yml, fly.toml, terraform/
  tests/               # unit/ integration/ media_golden/ e2e/
```

Core provider interface (Python):

```
class TranscriptionProvider(Protocol):
    name: str
    async def transcribe(self, audio_url: str, *, language: str | None,
                         vocabulary: list[str]) -> Transcript: ...

class Publisher(Protocol):
    destination: Destination
    async def preflight(self, conn: Connection, export: Export) ->
PreflightResult: ...  # limits, creator_info, account type
    async def publish(self, conn: Connection, export: Export, meta: PostMeta,
                      idempotency_key: str) -> PublishResult: ...
    async def status(self, conn: Connection, external_id: str) ->
PublishStatus: ...
```

Compose (local):

```
services:
  postgres: { image: pgvector/pgvector:pg16, environment: {
POSTGRES_PASSWORD: dev }, ports: ["5432:5432"] }
  redis:    { image: redis:7, ports: ["6379:6379"] }
  minio:    { image: minio/minio, command: server /data --console-address
:9001, ports: ["9000:9000","9001:9001"] }
  api:      { build: ., command: uvicorn apps.api.main:app --reload --host
0.0.0.0, env_file: .env, depends_on: [postgres, redis, minio], ports:
["8000:8000"] }
  worker:   { build: ., command: celery -A workers.app worker -Q
ingest,stt,analyze,render,publish -c 2, env_file: .env, depends_on: [api] }
  web:      { build: apps/web, command: npm run dev, ports: ["3000:3000"] }
```

Conventions: errors as problem+json with stable `code`; every domain error maps to one user message; tests named `test_<unit>_<behaviour>`; no PR merges without passing CI and one review. Placeholders: provider credentials, legal text, pricing values, scoring weights. Production-ready after configuration: schema, job framework, render pipeline, auth, billing webhooks — once the test gates above pass.

9.4 AI coding-agent work packages

Agree these contracts before parallel work: OpenAPI spec v1, database schema v1, render-spec JSON schema, job payload schemas, provider protocols, error-code list.

Package	Owns	Depends on	Deliverable / acceptance	Must not touch
WP1 Design system + web shell	apps/web/components, layout, auth pages	OpenAPI client	Storybook, axe clean	API, schema
WP2 Identity & workspaces	core/identity, core/workspaces, auth routers	Schema	RLS + cross-tenant tests pass	Media, billing
WP3 Media ingestion	core/media, uploads routers, ingest worker	WP2	Resumable 10 GB upload; ffprobe corpus passes	Render code
WP4 AI pipeline	core/transcripts, core/highlights, stt/llm providers	WP3 outputs	Precision@10 report, cost per hour logged	API contracts

Package	Owns	Depends on	Deliverable / acceptance	Must not touch
WP5 Editor + render	apps/web/editor, core/render, render worker	Render-spec schema	Golden media tests pass	Highlight scoring
WP6 Integrations	providers/twitch, youtube, tiktok, publishing module	WP2	Fakes + sandbox tests; idempotency proven	Billing
WP7 Billing	core/billing, webhooks	WP2	Stripe test-clock scenarios pass	Media
WP8 Infra & QA	infra/, CI, e2e, dashboards	All	One-command local up; staging deploy	Domain logic

Process: each agent works on a branch per package, opens a PR with tests; CI runs the full suite plus contract diff; a human (or reviewer agent with no write access) reviews against the package's "must not touch" list; integration happens daily on main behind feature flags.

9.5 Launch checklist

- [ ] TikTok and YouTube audits approved (else UI states "private only")
- [ ] Terms, privacy policy, DMCA agent, takedown form, cookie banner reviewed by counsel
- [ ] DPAs signed with STT, LLM, storage, email, payments vendors
- [ ] Pentest done, criticals fixed; cross-tenant suite green
- [ ] Backups restored successfully in staging
- [ ] Spend caps and cost alerts configured on every AI vendor
- [ ] Support inbox, status page, incident runbook, on-call rota
- [ ] Stripe + Paystack live, tax settings (VAT/GST) configured
- [ ] Analytics events and funnel dashboard live

9.6 Risks, assumptions and open questions

Risk	Likelihood	Impact	Mitigation
Highlight quality weak on low-speech gameplay	High	High	Chat + audio-event signals; label data per genre; honest "low confidence" UI
Audit rejected or delayed	Medium	High	Apply in week 1; download fallback; consider a vetted unified publishing API as interim
Twitch Clip-from-VOD beta changes	Medium	Medium	Feature flag; uploads remain primary
Incumbents ship stream features	High	Medium	Move fast on editor authorization + per-creator learning
Free-tier abuse	Medium	Medium	Low free allowance, verification, device/card heuristics
Copyright complaints (music, third-party content)	Medium	High	Rights model, takedowns, music warnings, repeat-infringer policy
AI cost spikes	Low	Medium	Spend caps, vendor failover, self-host path

Open questions: exact Twitch scopes for Get Clips Download and Clip-from-VOD (confirm in Helix reference); YouTube upload quota bucket in your own Cloud Console; Instagram daily cap (docs state both 50 and 100); Kick VOD API plans; Paystack fees and NGN pricing; whether streamers will let a bot record chat.

9.7 Sources (accessed 17 Oct 2026)

- Twitch Clips guide and Clip-from-VOD open beta announcement
- Twitch forum: VOD downloads not supported
- Kick.NET SDK notes on official vs undocumented endpoints; kick-api Rust crate
- TikTok Content Sharing Guidelines
- Instagram Content Publishing
- YouTube upload quota and audit: PostProxy, singhamandeep.com — confirm against Google's quota table

- X pricing: PostProxy, OpenTweet — confirm at docs.x.com
- OpusClip pricing summary (Creatify)
- STT pricing: convertaudiototext, Deepgram Nova-3 explainer

Several figures come from third-party summaries rather than vendor pages; each is marked "verify" where it matters for a build or pricing decision.